

ENTERING EXPRESSIONS

Type and press **ENTER**. **Ans** is the previous result. **DEL** backspaces; **`** leaves. **[]** scroll the tape · **'** toggles the on-screen hints · **** engineering notation · **F_n+p** prints the tape · **F_n+f** full function list.

SI SUFFIXES

Any number takes **p n u m k M G T** — **145M** is 145 000 000. Mixing them is fine:
fspl(145M/1M, 800).

ARITHMETIC

+ - * / ^ () mod · sqrt cbt abs sign round floor ceil · min(a,b) max(a,b) hypot(a,b) · fact(n) ncr(n,r) npr(n,r).

LOGS AND POWERS

ln natural · **log** base 10 · **log2** · **exp**.

TRIGONOMETRY — DEGREES

sin cos tan asin acos atan take and return **degrees**, not radians. **atan2(y,x)** keeps the quadrant — use it for bearings. **sinh cosh tanh** · **d2r r2d** convert.

CONSTANTS

pi e · **c** speed of light m/s · **kB** Boltzmann J/K · **Re** Earth radius 6378.137 km · **mu** Earth GM 398600.4418 · **g0** 9.80665.

GRAPHER

Same language, plus **x** as the plotted variable. **ENTER** edits **y=f(x)**, **2** a second trace, **t** trace, **m** mark, **z** zero, **b** table, **csv** export. **F_n+f** opens this list.

DECIBELS AND POWER

db(x) ratio→dB · **undb(x)** dB→ratio · **dbm(w)**
 W→dBm · **w(dbm)** dBm→W · **dbm2w** /
w2dbm the same pair by the other name ·
dbd(dbi) and **dbi(dbd)** shift antenna gain by
 2.15 dB.

SWR, FEEDLINE, NOISE

swr2rl(swr) and **rl2swr(db)** · **mml(swr)**
 mismatch loss dB · **nf2t(nf)** noise
 figure→kelvin · **t2nf(k)** back again.

FREQUENCY AND ANTENNAS

wl(mhz) / **fq(m)** and **lam(mhz)** wavelength ·
dipole(mhz) half-wave length in metres (0.95
 velocity factor, so it is the length to cut) ·
dgain(d,mhz) parabolic dish dBi at 55%
 efficiency · **fspl(mhz,km)** free-space path loss
 dB.

SATELLITE

porb(alt) orbital period in minutes · **aorb(min)**
 the altitude that gives a period · **vorb(alt)**
 orbital speed km/s · **fpr(alt)** footprint radius km
 · **slant(el,alt)** actual range at that elevation ·
dop(mhz,rr) Doppler Hz from range rate km/s.

Altitudes in km.

WORKED EXAMPLES

lam(435) → 0.689 m · **dipole(14.1)** → 10.10 m
 · **porb(420)** → 93.0 min · **slant(0,420)** → 2352
 km at the horizon, against 420 km overhead —
 the reason to use it · **fspl(435,2352)** → 152.6
 dB · **dgain(3,10368)** → 47.7 dBi.

WHY SOME OF THESE EXIST

atan2 because a bearing from **atan(y/x)** loses
 the quadrant and divides by zero due east.
slant because treating altitude as range
 understates a horizon path by 5.6x, exactly
 where the link budget is tightest. **dipole**
 because a free-space half wave cuts elements
 measurably long.